

multimodal-disagreement: A Python package for quantifying inter-modal disagreement in deep learning exoplanet classifiers

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Software

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Summary

Multimodal deep learning architectures are increasingly the standard for automated exoplanet detection in large-scale transit surveys, integrating 1D time-series light curves and 2D spatial centroid representations. The multimodal-disagreement Python package provides tools for computing inter-modal disagreement metrics in these multimodal exoplanet classification pipelines. By applying structured masking to isolate modality-specific predictions, the software enables researchers to compute an inter-modal disagreement metric (δ) and implement a computationally inexpensive “Disagreement Veto” ($\delta > 0.3$) to filter out false-positive transit signals in automated pipelines.

Statement of need

The Kepler space telescope identified thousands of planetary candidates by monitoring the brightness of over 150,000 stars (Borucki & others, 2010). As dataset scales increased, machine learning methods—particularly deep neural networks—were introduced to improve transit classification accuracy, establishing new benchmarks for mission data releases (Dattilo & others, 2019; Shallue & Vanderburg, 2018).

Existing machine learning pipelines typically provide only fused prediction probabilities and do not expose modality-specific diagnostics. This makes it difficult to evaluate internal model consistency when confronted with sparse or ambiguous transit signals. This package provides tools for extracting modality-specific predictions and computing disagreement metrics that can be used to analyze model stability.

State of the field

While there are several robust open-source packages for exoplanet transit modeling and probabilistic inference (such as exoplanet and kima), these tools generally focus on parameter estimation and signal fitting. Deep learning pipelines often use late-fusion architectures to combine multiple data representations, but they lack built-in tools to evaluate whether the temporal and spatial branches actually agree with one another. multimodal-disagreement fills this specific niche by providing a post-hoc diagnostic framework to audit the internal consistency of existing deep learning classifiers without requiring independent retraining of unimodal networks.

Software design

The multimodal-disagreement package is implemented in Python and organized into modular components for prediction extraction, disagreement analysis, and diagnostic visualization.

Key functionality includes: - Extraction of modality-specific predictions using structured masking. - Computation of the disagreement metric $\delta = |P_{temporal} - P_{spatial}|$. - Utilities for analyzing disagreement across astrophysical parameters (such as orbital period regimes). - Command-line tools and batch scripts for large-scale survey dataset analysis.

The software is designed to be framework-agnostic at the extraction layer, meaning it can be integrated into existing TensorFlow/Keras or PyTorch classification pipelines or used independently for post-hoc diagnostic analysis.

Research impact statement

Automated vetting pipelines for missions like TESS and the upcoming PLATO mission must process millions of light curves, making human review impossible. By providing a computationally inexpensive metric to identify internal model disagreement, this software allows researchers to flag systematic failure modes (such as “reliability cliffs” at specific orbital periods) and systematically reduce false-positive rates. This improves the purity of exoplanet candidate catalogs generated by deep learning.

Installation

The package can be installed directly from source:

```
git clone [https://github.com/](https://github.com/)<username>/multimodal-disagreement
cd multimodal-disagreement
pip install -e .
```

Usage Example

The package provides utilities for extracting modality-specific predictions from multimodal classifiers and computing the inter-modal disagreement metric. A minimal example demonstrating the calculation of the disagreement score (δ) and the application of the veto is shown below:

```
from multimodal_disagreement.metrics import disagreement, apply_veto
```

```
# Simulated independent branch probabilities
```

```
p_temporal = 0.91
```

```
p_spatial = 0.12
```

```
# Calculate the disagreement metric (delta)
```

```
delta = disagreement(p_temporal, p_spatial)
```

```
print(f"Inter-modal Disagreement: {delta}")
```

```
# Output: Inter-modal Disagreement: 0.79
```

```
# Apply the recommended Disagreement Veto threshold
```

```
is_vetoed = apply_veto(delta, threshold=0.3)
```

```
print(f"Flagged as False Positive: {is_vetoed}")
```

```
# Output: Flagged as False Positive: True
```

59 **AI usage disclosure**

60 Artificial intelligence tools, specifically large language models, were used during the preparation
61 of this manuscript for copyediting, formatting, and structural organization to adhere to the
62 Journal of Open Source Software guidelines.

63 **Acknowledgements**

64 The author confirms sole responsibility for software architecture, implementation, and
65 documentation. Data supporting the validation of this software can be found in the Kepler
66 Data Release 25 archive at the NASA Exoplanet Archive ([NASA Exoplanet Science Institute, 2026](#)).
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68 **References**

- 69 Borucki, W. J., & others. (2010). Kepler planet-detection mission: Introduction and first
70 results. *Science*, 327, 977–980. <https://doi.org/10.1126/science.1185402>
- 71 Dattilo, A., & others. (2019). Identifying exoplanets with deep learning. II. Two new super-
72 earths uncovered by a neural network in K2 data. *The Astronomical Journal*, 157(5), 169.
73 <https://doi.org/10.3847/1538-3881/ab0e12>
- 74 NASA Exoplanet Science Institute. (2026). *NASA exoplanet archive*. [https://exoplanetarchive.
75 ipac.caltech.edu](https://exoplanetarchive.ipac.caltech.edu)
- 76 Shallue, C. J., & Vanderburg, A. (2018). Identifying exoplanets with deep learning: A
77 five-planet resonant chain around kepler-80 and an eighth planet around kepler-90. *The
78 Astronomical Journal*, 155(2), 94. <https://doi.org/10.3847/1538-3881/aa9e09>